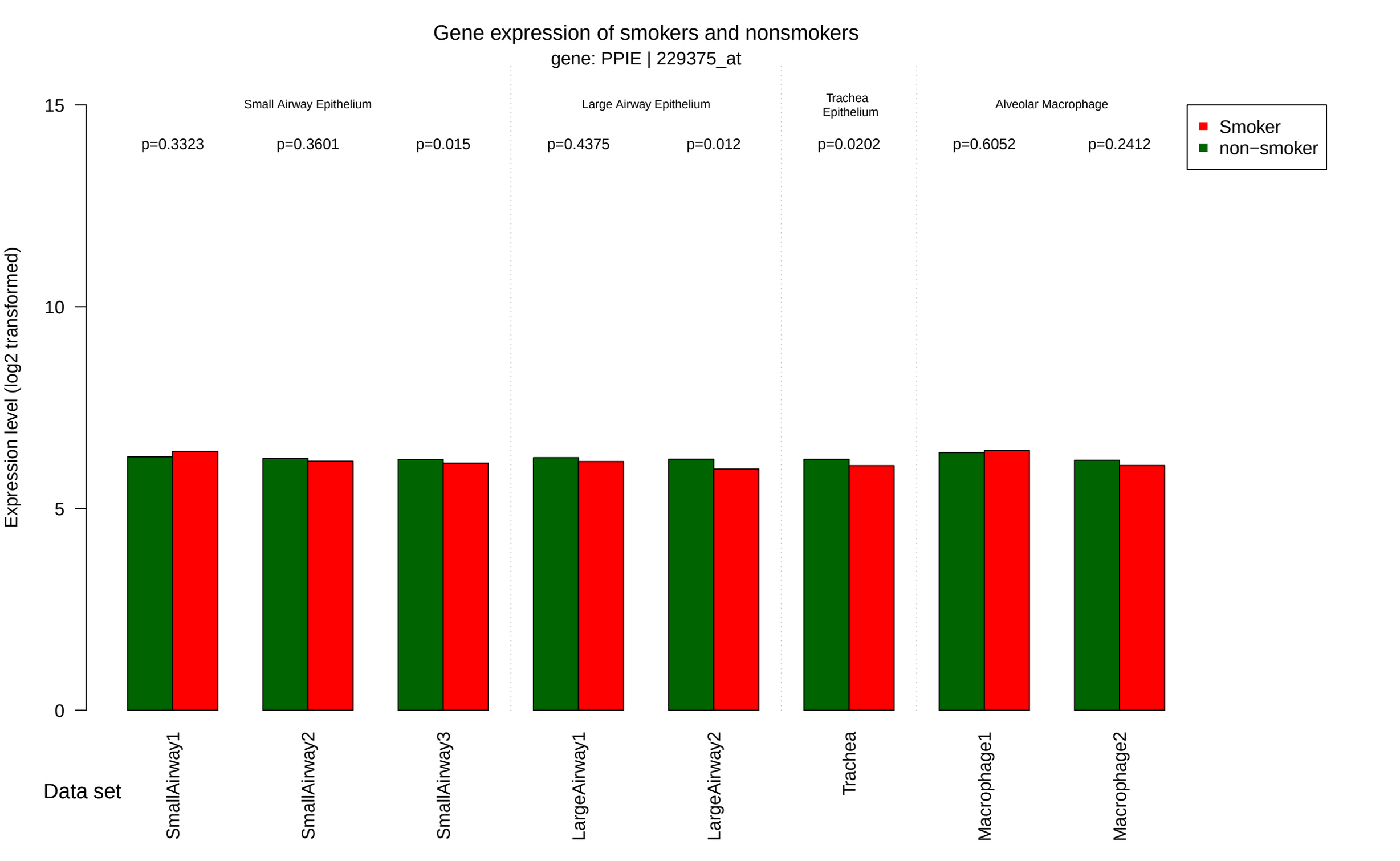
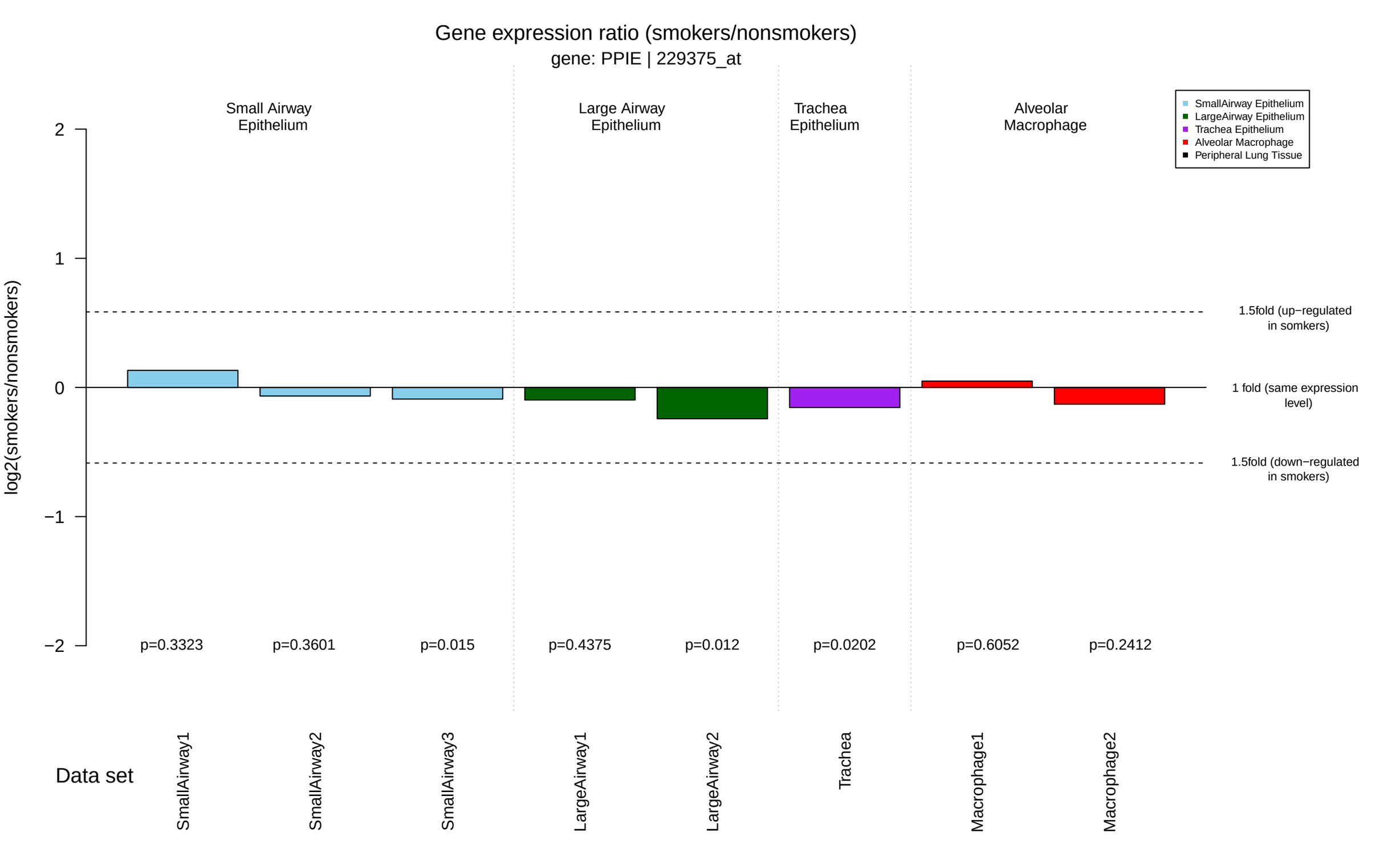
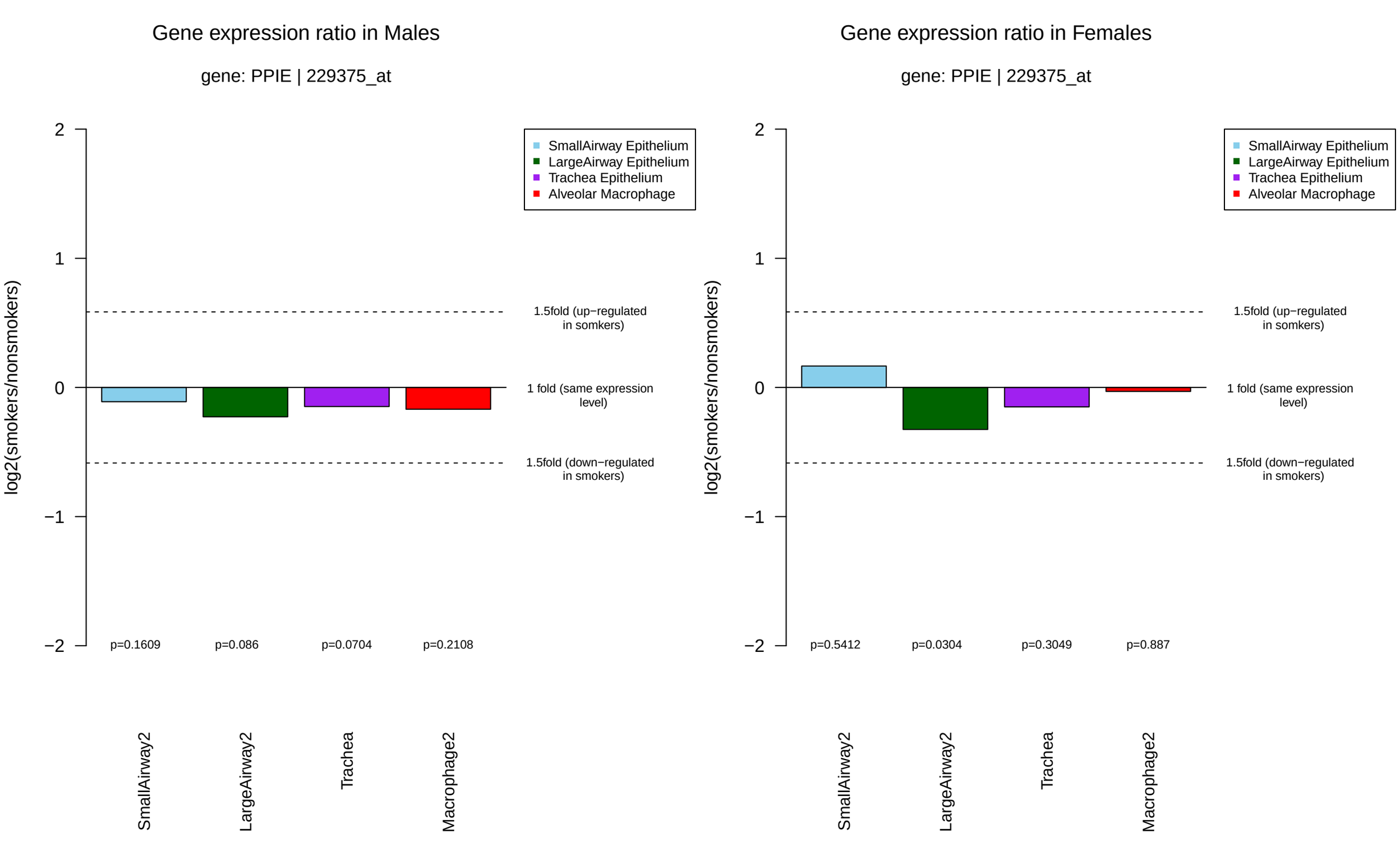
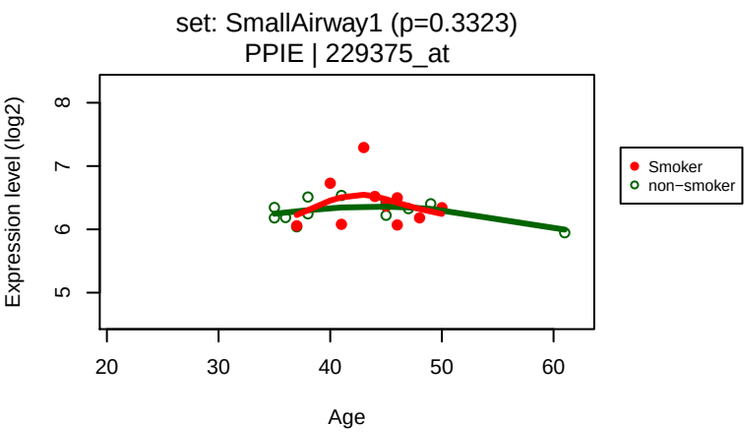
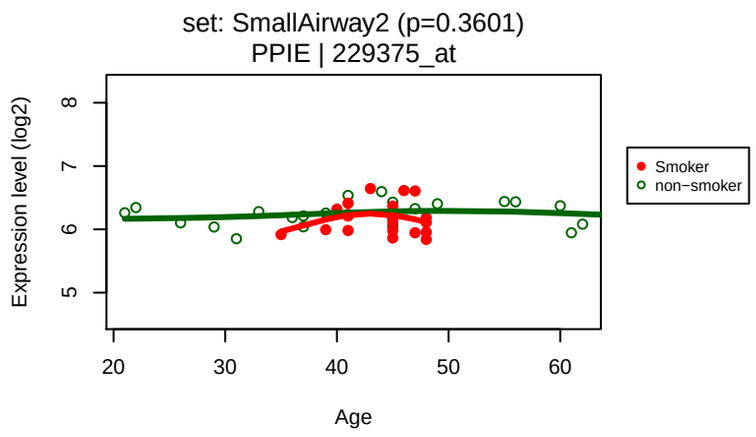
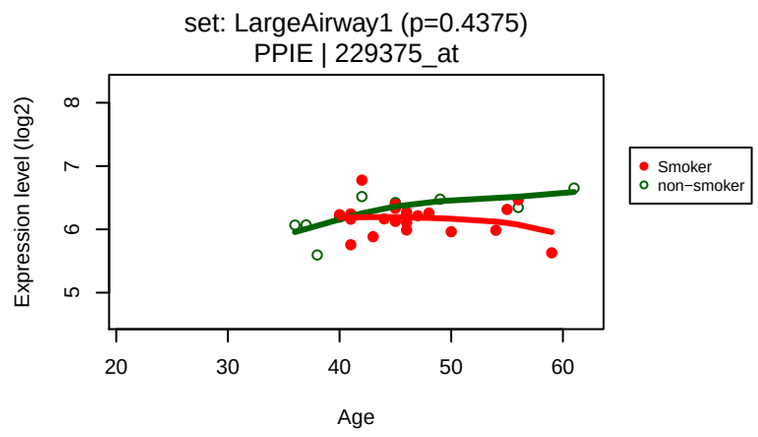

Gene expression of smokers and nonsmokers
gene: PPIE | 229375_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3323
p=0.3601
p=0.015
p=0.4375
p=0.012
p=0.0202
p=0.6052
p=0.2412
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

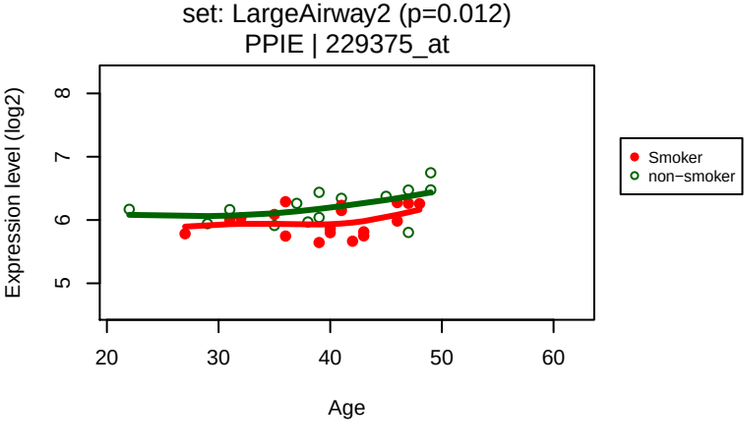

Gene expression ratio (smokers/nonsmokers)
gene: PPIE | 229375_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3323
p=0.3601
p=0.015
p=0.4375
p=0.012
p=0.0202
p=0.6052
p=0.2412
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

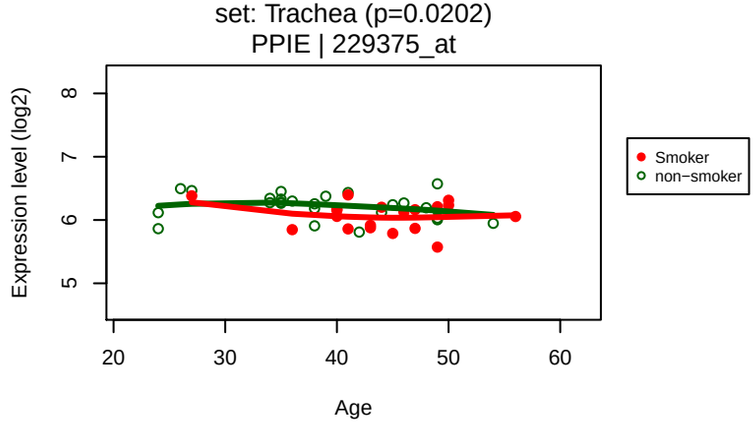

Gene expression ratio in Males
gene: PPIE | 229375_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1609
p=0.086
p=0.0704
p=0.2108
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPIE | 229375_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5412
p=0.0304
p=0.3049
p=0.887
SmallAirway2
LargeAirway2
Trachea
Macrophage2

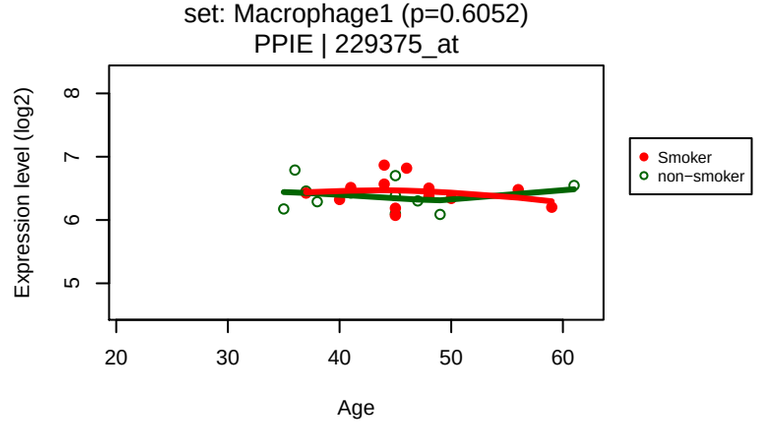

set: SmallAirway1 (p=0.3323)
PPIE | 229375_at
Expression level (log2)
Age
Smoker
non−smoker

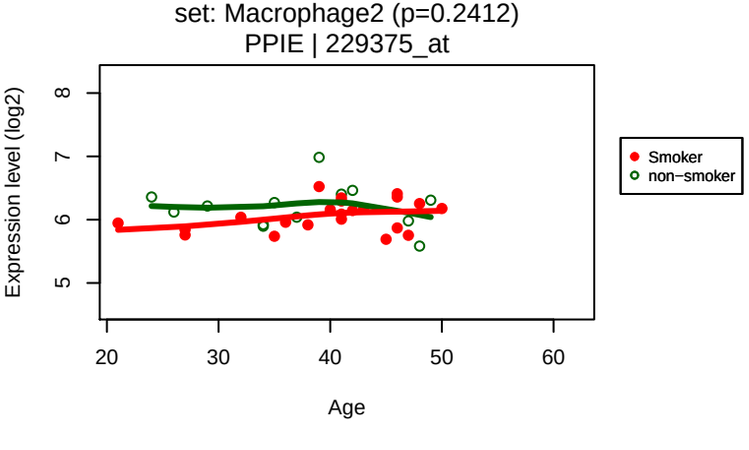

set: SmallAirway2 (p=0.3601)
PPIE | 229375_at
Expression level (log2)
Age
Smoker
non−smoker

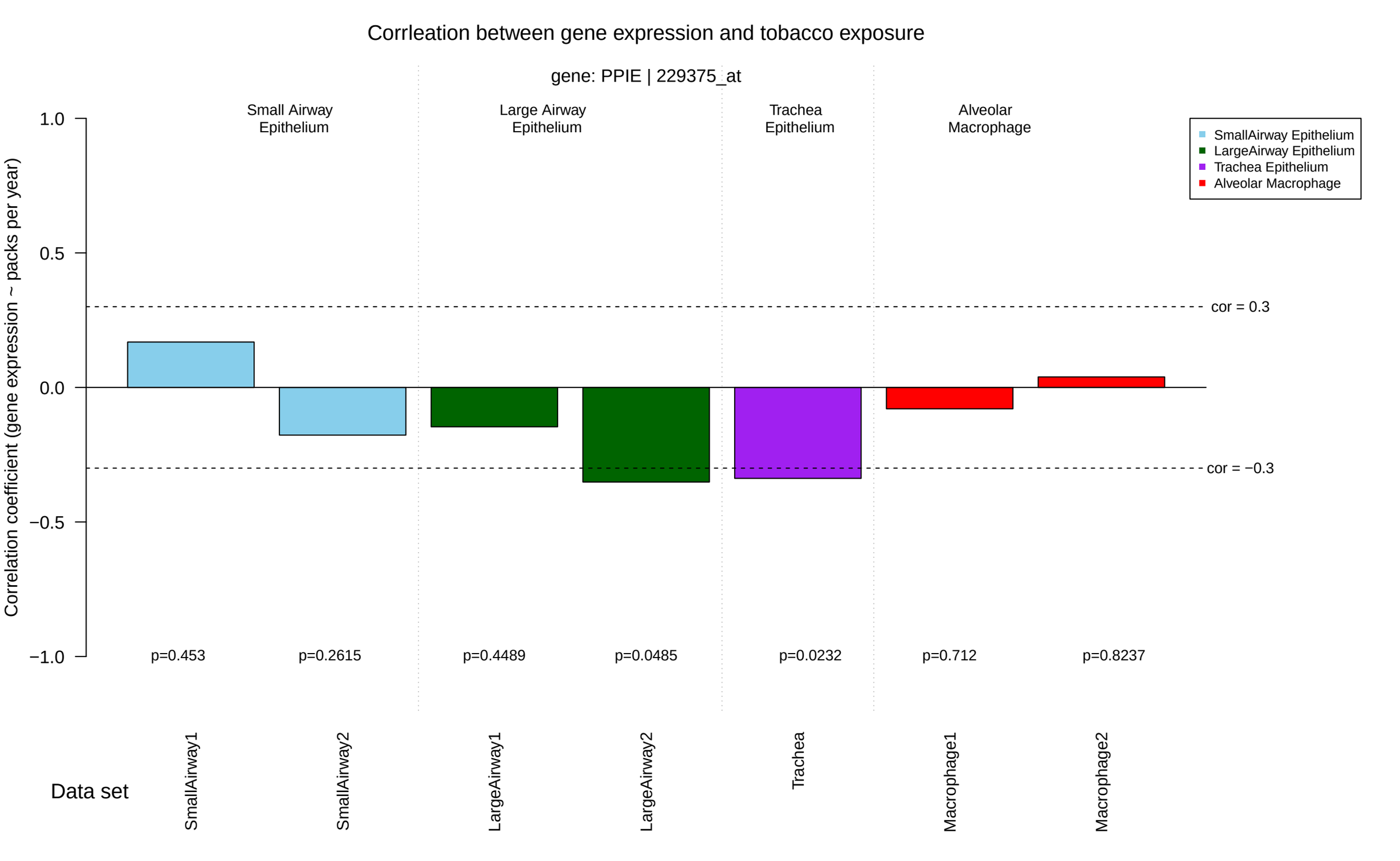

set: LargeAirway1 (p=0.4375)
PPIE | 229375_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.012)
PPIE | 229375_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0202)
PPIE | 229375_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6052)
PPIE | 229375_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2412)
PPIE | 229375_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPIE | 229375_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.453
p=0.2615
p=0.4489
p=0.0485
p=0.0232
p=0.712
p=0.8237
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2